

Skilog: A Smart Sensor System for Performance Analysis and Biofeedback in Ski Jumping

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1 Introduction

In ski jumping, **low repetition rates** of jumps **limit** the effectiveness of **training**. Thus, increasing learning rate within every single jump is key to success.

A critical element of athlete training is **motor learning**, which has been shown to be accelerated by bio-feedback methods. A fine-grained control of the center of gravity in the in-run is essential. This is because the actual takeoff occurs within a blink of an eye (approx. 300 ms), thus any unbalanced body posture during the in-run will affect flight.

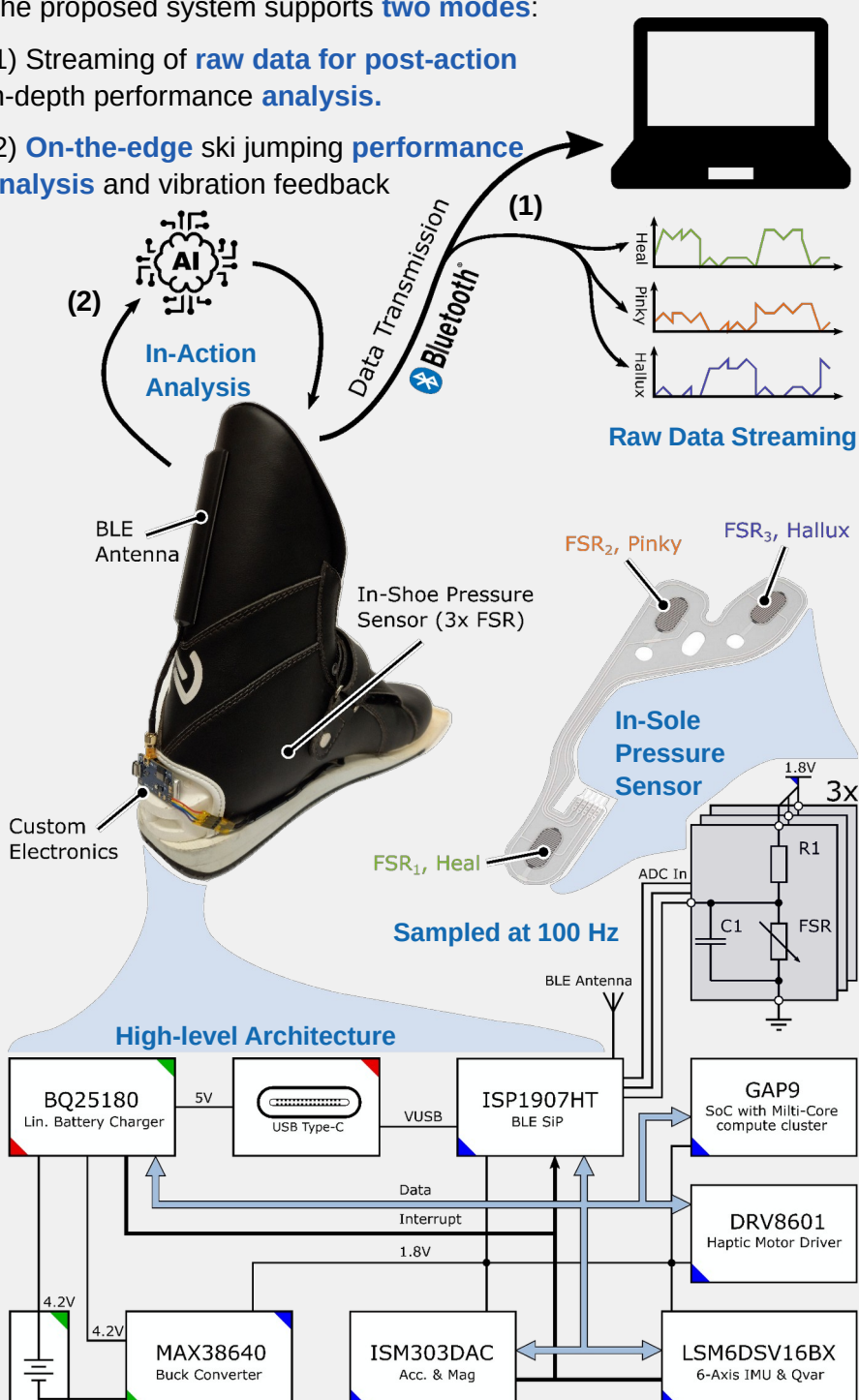
This work presents a smart, compact, and energy-efficient wireless sensor system for **real-time performance analysis** and **biofeedback** during ski jumping.

2 System Definition

The proposed system supports **two modes**:

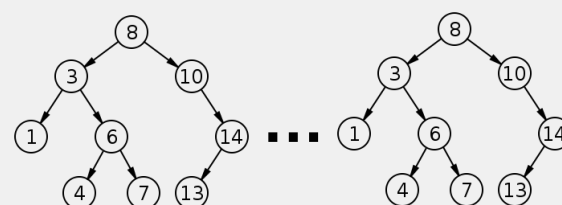
(1) Streaming of **raw data for post-action** in-depth performance **analysis**.

(2) **On-the-edge** ski jumping **performance analysis** and vibration feedback



3 In-Action Analysis

We employ **XGBoost** for classification, leveraging its **Extreme Gradient Boosting** algorithm. Each of the **three classes** (neutral, ventral and dorsal position) is represented by a forest of decision trees. During inference, these trees are traversed to deduce the type of action performed by the jumper.



Typical forest of decision trees in XGBoost.

Inference is **lightning fast** (0.011ms)

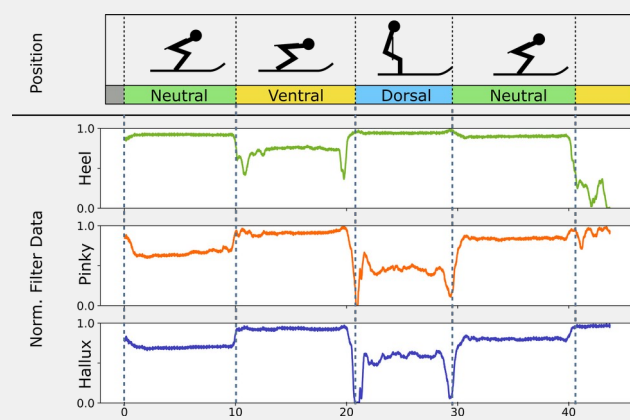


and **energy-efficient** (0.251 μ J)



The XGBoost trees are stored in a **memory-optimized format**, requiring merely **7 bytes per node**, with the entire model occupying just **38.29 kB**.

4 Experimental Results

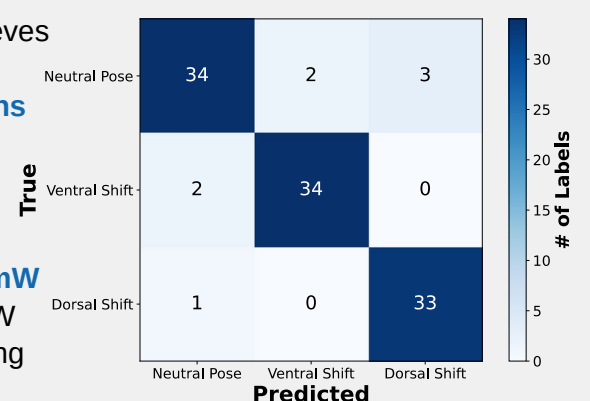


Normalized raw data collected under laboratory conditions.

Body positions were estimated from **raw pressure sensor data** captured at 3 measurement points in the ski boot (Pinky, Hallux, Heel). Data was **sampled at 100 Hz**. Neutral position were determined by the centroid of the three sensing points.

The XGBoost model achieves **92.7 % accuracy in identifying body positions** from laboratory-recorded data.

Our systems' **total power consumption is at 2.52 mW** (848 μ W data Rx, 1.67 mW data Tx over BLE), enabling multiday battery lifetime support.



Confusion matrix of the deployed XGBoost model.

4 Conclusion

This paper presented a **novel smart sensing and training system for body position analyses during ski jumping**. Our system can Tx data to coaches and provide direct **in-action feedback** to athletes. It is the first truly wearable training tool in ski jumping that can provide ski jumpers with a new training experience and **could shorten the time for motor learning**. Moreover, real-time data transmission of biomechanical relevant features can impact TV broadcasting.